

Requirements Engineering & UML Use-Case Modelling

Project Report

Problem Statement #06 — Campus & Academic Operations

Alumni Mentorship & Mock Interview Platform

Project	Requirements Engineering & UML – Case Modelling
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Course / Lab	Software Engineering — Lab 1

1 Problem Context

Connecting students with alumni working in specialized domains requires an intelligent matching mechanism, an availability calendar-booking flow, and structured post-interview scorecard recording. The platform's target stakeholders / actors are the Student Mentee and the Alumni Mentor.

2 Requirements Table

Five functional requirements (FR-001–FR-005) and two non-functional requirements (NFR-001–NFR-002), each with ID, description, priority, measurable acceptance criteria, and rationale.

2.1 Functional Requirements

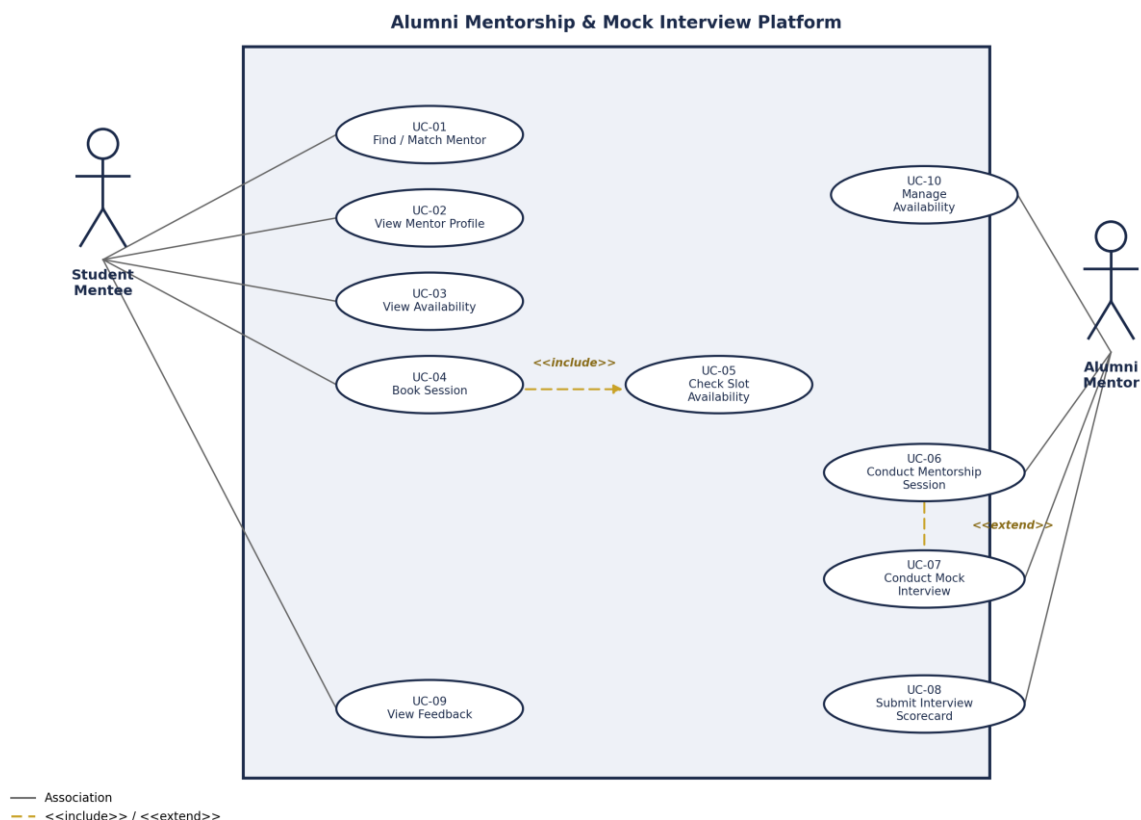
FR-001	Recommend alumni mentors to students based on domain matching (e.g., Embedded Systems, Cloud Computing) and facilitate one-click session booking.	High	Pass: Mentorship session is added to both calendars with a meeting link. Fail: Double-booking of mentor slots.	Core matching function that anchors the platform's value proposition.
FR-002	Display an Alumni Mentor's available time slots to a Student Mentee before booking.	High	Pass: Displayed slots reflect the mentor's live calendar at time of viewing. Fail: A slot shown as free is already booked.	Prevents wasted booking attempts and gives students accurate scheduling information.
FR-003	Allow a Student Mentee to book an available mentorship or mock-interview session with a selected Alumni Mentor.	High	Pass: Session is created, added to both calendars, and a meeting link is generated. Fail: Booking of an already-reserved slot succeeds (double-booking).	Central transactional workflow of the platform; directly tied to the assignment's sample acceptance criteria.
FR-004	Allow an Alumni Mentor to add, modify, or remove their available time slots.	Medium	Pass: Updated availability is reflected to students within the same session. Fail: Students continue to see stale/removed slots.	Keeps supply-side availability accurate and avoids downstream booking conflicts.
FR-005	Allow an Alumni Mentor to record and submit a structured post-interview scorecard for a completed session.	Medium	Pass: Scorecard is saved, linked to the correct session, and visible to the student. Fail: Scorecard is lost or linked to the wrong session.	Fulfills the problem statement's requirement for structured post-interview feedback.

2.2 Non-Functional Requirements

NFR-001	Privacy & Security	All user profile data and feedback scorecards shall adhere to privacy guidelines and support soft-deletion within 24 hours of a deletion request.	High	Pass: Benchmarking/audit tests confirm target latency and security standards under simulated peak load, and soft-deleted records are inaccessible within 24 hours. Fail: Deleted data remains queryable after 24 hours.	Protects sensitive student/mentor data and satisfies data-protection expectations for an institutional platform.
NFR-002	Performance	The system shall return mentor search/recommendation results within 2 seconds for 95% of requests under normal load.	Medium	Pass: Load testing confirms sub-2-second response time at the 95th percentile. Fail: 95th-percentile response time exceeds 2 seconds.	Maintains a responsive experience for students, particularly during peak placement/interview seasons.

3 UML Use-Case Diagram

Two primary actors were identified directly from the problem statement — Student Mentee and Alumni Mentor. Ten use cases model the full interaction surface, including one supporting use case (Check Slot Availability) and one alternative-behaviour use case (Conduct Mock Interview), which realise the required « include » and « extend » relationships respectively.



3.1 Actor → Use-Case Mapping

UC-01	Find / Match Mentor	Student Mentee
UC-02	View Mentor Profile	Student Mentee
UC-03	View Availability	Student Mentee
UC-04	Book Session	Student Mentee
UC-05	Check Slot Availability	(supporting — included by UC-04)
UC-06	Conduct Mentorship Session	Alumni Mentor
UC-07	Conduct Mock Interview	Alumni Mentor
UC-08	Submit Interview Scorecard	Alumni Mentor
UC-09	View Feedback	Student Mentee
UC-10	Manage Availability	Alumni Mentor

3.2 Relationships

« include » — Book Session → Check Slot Availability

Booking a session always requires checking whether the selected mentor slot is still available, so this is mandatory, reusable behaviour — the textbook case for « include ».

« extend » — Conduct Mock Interview → Conduct Mentorship Session

A mentorship session does not always involve interview practice; a mock interview is an optional specialisation that extends the base session only when the student requests it, so this is conditional behaviour — the correct case for « extend ».

4 Use-Case Flow Specification

One key use case — Book Mentorship / Mock Interview Session — is specified below with preconditions, the main success scenario, one alternate flow, and postconditions.

Use Case ID	UC-04
Use Case Name	Book Mentorship / Mock Interview Session
Primary Actor	Student Mentee
Supporting Actor	Alumni Mentor
Priority	High

4.1 Goal

To allow a Student Mentee to reserve an available mentorship or mock-interview session with a selected Alumni Mentor.

4.2 Preconditions

- The Student Mentee is registered and authenticated on the platform.
- The student has selected an Alumni Mentor.
- The selected mentor has at least one available time slot.
- The student has selected the required session type (mentorship or mock interview).

4.3 Main Success Scenario

1. The Student Mentee selects an Alumni Mentor from the recommended mentor list.
2. The system displays the mentor's available time slots.
3. The Student Mentee selects an available time slot.
4. The Student Mentee selects the session type and requests the booking.
5. The system checks whether the selected time slot is still available.
6. The system reserves the selected slot for the student.
7. The system creates the mentorship / mock-interview session.
8. The system adds the session to both the student's and mentor's calendars.
9. The system generates or associates a meeting link with the session.
10. The system displays a booking confirmation to the Student Mentee.

4.4 Alternate Flow A1 — Selected Slot Is Unavailable

Branches from Step 5 of the main success scenario:

1. The system determines that the selected slot has already been booked or is no longer available.
2. The system rejects the booking request.
3. The system informs the Student Mentee that the selected slot is unavailable.
4. The system displays the mentor's currently available slots.
5. The Student Mentee selects another available slot and the flow resumes from Step 4 of the main success scenario.

4.5 Postconditions

Successful completion:

- A mentorship or mock-interview session is created for the student and the selected alumni mentor.
- The selected time slot is marked unavailable for further booking.
- The session appears on both the student's and mentor's calendars.
- A meeting link is associated with the session.
- The student receives a booking confirmation.

Alternate completion:

- No duplicate booking is created, and the selected slot remains unavailable.

USE-CASE FLOW SPECIFICATION

UC-01 — Intelligent Mentor Matching

Primary Actor: Student Mentee

Supporting Actor: Alumni Mentor

Goal: Recommend suitable alumni mentors based on the student's selected domain and requirements.

Preconditions

1. Student Mentee is registered and authenticated.
2. Student has selected an area of interest.
3. Alumni mentor profiles are available in the system.
4. Mentor profiles contain domain/expertise information.

Main Success Scenario

1. Student selects **Find a Mentor**.
2. System displays available domains/interests.
3. Student selects a desired domain.
4. System retrieves alumni profiles matching the selected domain.
5. System evaluates mentor profiles based on relevant expertise.
6. System ranks matching mentors according to relevance.
7. System displays recommended mentors.
8. Student selects a recommended mentor.
9. System displays the mentor's profile and available session options.
10. Student proceeds to view availability and book a session.

Alternative Flow A1 — No Matching Mentor**At Step 4:**

1. System finds no alumni mentor matching the selected domain.
2. System informs the student that no suitable mentor is currently available.
3. System suggests related domains or broader matching criteria.
4. Student selects an alternative criterion.
5. System repeats the matching process from Step 4.

Alternative Flow A2 — Insufficient Student Information**At Step 3:**

1. System determines that insufficient information is available to perform effective matching.
2. System requests additional domain or interest information.
3. Student provides the required information.
4. System resumes the matching process from Step 4.

Alternative Flow A3 — Recommended Mentor Becomes Unavailable**At Step 8:**

1. System detects that the selected mentor is no longer available.
2. System informs the student.
3. System refreshes the recommended mentor list.
4. Student selects another recommended mentor.

Exception Flow E1 — Matching Service Failure**At Step 4:**

1. The mentor-matching service fails to respond.

2. System displays an error message.
3. System does not display unverified recommendations.
4. Student can retry the matching request.

Postconditions**Success:**

- Suitable mentors are displayed.
- Recommendations are ranked.
- Student can proceed to mentor profile and booking.

Failure:

- No invalid recommendation is displayed.
- Student receives an appropriate message.
- Student can retry or modify their search criteria.